

EX.NO.8b	JAVASCRIPT DOM (REAL-TIME DIGITAL CLOCK)
DATE: 07.09.26	

AIM:

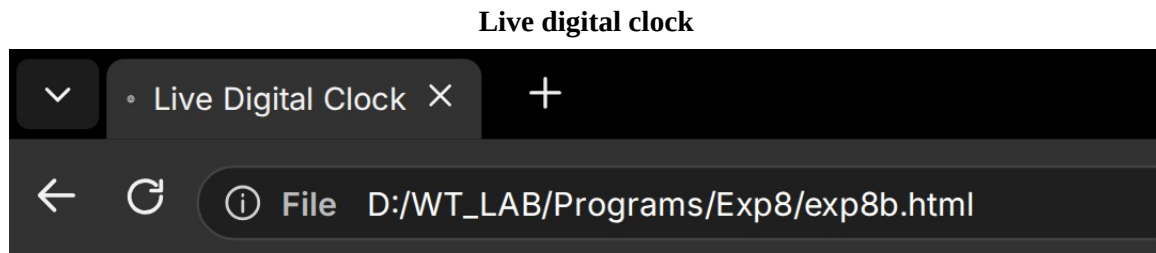
To create a digital clock web page that shows the current date and the live time in 12-hour format using JavaScript timers.

ALGORITHM:

1. Create a page with two elements, one for the date and one for the time.
2. Write a function that reads the current Date object and formats hours, minutes and seconds.
3. Convert the hours to the 12-hour format and add the AM or PM label.
4. Pad single-digit values with a leading zero.
5. Call the function once and then repeat it every second with setInterval().
6. Run the page and confirm that the time changes every second.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Live Digital Clock</title>
  <style>
    body { font-family: Arial; text-align: center; margin-top: 70px; }
    #date { font-size: 20px; color: #444; }
    #time { font-size: 56px; font-weight: bold; color: #0b3d91; letter-spacing: 3px; }
  </style>
</head>
<body>
  <h2>Live Digital Clock</h2>
  <div id="date"></div>
  <div id="time"></div>
  <script>
    function two(n) { return n < 10 ? "0" + n : n; }
    function tick() {
      var d = new Date();
      var h = d.getHours();
      var ap = h >= 12 ? "PM" : "AM";
      h = h % 12 || 12;
      document.getElementById("time").innerHTML =
        two(h) + ":" + two(d.getMinutes()) + ":" + two(d.getSeconds()) + " " + ap;
      document.getElementById("date").innerHTML = d.toString();
    }
    tick();
    setInterval(tick, 1000);
  </script>
</body>
</html>
```

OUTPUT:**RESULT:**

Thus, a real-time digital clock was created using JavaScript and it updates every second.